WORKED SOLUTION

STEP III 2004, Question 4

Pure • Euclidean geometry

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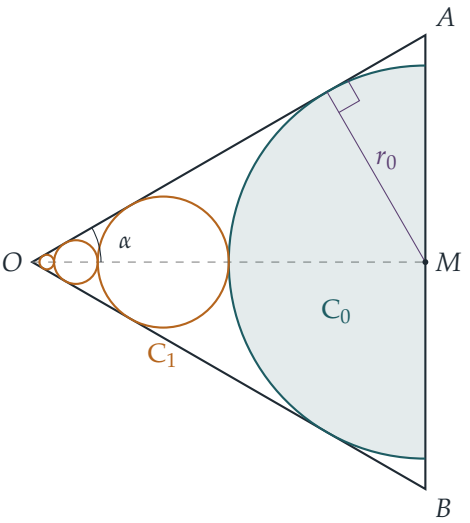
Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/04-s3-q4

The key idea

Every circle that touches both OA and OB has its centre on the bisector of angle AOB, and its distance from O is fixed by its radius: a right-angled triangle with angle α at O gives distance $= r/\sin\alpha$. Writing "the two circles touch" as an equation in these distances links consecutive radii, and the radii form a geometric progression. The areas then form a geometric series, and the last part is a one-variable maximisation (or a well-chosen value of α).

Solution



The ratio of radii

Let M be the midpoint of AB. Since $OA = OB$, the line OM bisects angle AOB and is perpendicular to AB; each of the angles AOM and BOM is α.

Take any circle of radius r touching both lines OA and OB. Its centre is equidistant from the two lines, so it lies on the bisector OM. If the centre is P and the circle touches OA at T, then $PT \perp OA$, $PT = r$ and angle $POT = \alpha$, so in the right-angled triangle OTP

$$OP = \frac{r}{\sin\alpha}.$$

This applies to C_0 (centre M) and to every C_n. Write $d_n = r_n/\sin\alpha$ for the distance from O to the centre of C_n.

The circle C_{n+1} sits in the corner between C_n and O (it must lie inside the angle, on the O side of C_n, since it touches the sides OA and OB, not their extensions). Two circles touching externally have centres a distance $r_n + r_{n+1}$ apart, and here both centres lie on OM, so

$$d_n - d_{n+1} = r_n + r_{n+1}.$$

Substituting $d_n = r_n/\sin\alpha$ and multiplying by $\sin\alpha$:

$$\begin{aligned} r_n - r_{n+1} &= (r_n + r_{n+1})\sin\alpha \\ r_n(1-\sin\alpha) &= r_{n+1}(1+\sin\alpha), \end{aligned}$$

and therefore

$$\frac{r_{n+1}}{r_n} = \frac{1-\sin\alpha}{1+\sin\alpha},$$

as required. This works for $n = 0$ too: C_1 touches the curved part of the semicircle, which is the half of C_0 facing O.

The total area

Write $k = \dfrac{1-\sin\alpha}{1+\sin\alpha}$. Since $0 < \sin\alpha < 1$, we have $0 < k < 1$, and $r_n = k^n r_0$. The semicircle has area $\frac{1}{2}\pi r_0^2$ and C_n has area $\pi k^{2n} r_0^2$. The series $\sum_{n\geq 1} k^{2n}$ is geometric with ratio $k^2 < 1$, so it converges to $k^2/(1-k^2)$. Thus

$$S = \pi r_0^2 \left(\frac{1}{2} + \frac{k^2}{1-k^2}\right).$$

Now

$$1 - k^2 = \frac{(1+\sin\alpha)^2 - (1-\sin\alpha)^2}{(1+\sin\alpha)^2} = \frac{4\sin\alpha}{(1+\sin\alpha)^2}, \qquad \text{so} \qquad \frac{k^2}{1-k^2} = \frac{(1-\sin\alpha)^2}{4\sin\alpha}.$$

Hence

$$S = \pi r_0^2 \, \frac{2\sin\alpha + 1 - 2\sin\alpha + \sin^2\alpha}{4\sin\alpha} = \frac{1+\sin^2\alpha}{4\sin\alpha}\,\pi r_0^2,$$

which is the result we wanted.

Comparing with the triangle

From the first part, $OM = d_0 = r_0/\sin\alpha$. In the right-angled triangle OMA, $AM = OM\tan\alpha$, so the area of triangle OAB is

$$T = OM \cdot AM = OM^2 \tan\alpha = \frac{r_0^2}{\sin^2\alpha} \cdot \frac{\sin\alpha}{\cos\alpha} = \frac{r_0^2}{\sin\alpha\cos\alpha}.$$

(The semicircle does fit on the base: $r_0 = OM\sin\alpha < OM\tan\alpha = AM$.) Dividing,

$$\frac{S}{T} = \frac{\pi}{4}(1+\sin^2\alpha)\cos\alpha.$$

Ask yourself Which value of α should we try?

As $\alpha \to 0$ the ratio tends to $\pi/4 \approx 0.785$, just short of 0.8, so very thin triangles don't work. A natural first try is a right angle at O, that is $\alpha = \pi/4$. Then $\sin^2\alpha = \frac{1}{2}$ and $\cos\alpha = 1/\sqrt{2}$, so

$$\frac{S}{T} = \frac{\pi}{4} \cdot \frac{3}{2} \cdot \frac{1}{\sqrt{2}} = \frac{3\pi}{8\sqrt{2}}.$$

This exceeds $\frac{4}{5}$ exactly when $15\pi > 32\sqrt{2}$. Both sides are positive, so we may square: the condition becomes $225\pi^2 > 2048$, that is $\pi^2 > 9.102\ldots$. Since $\pi > 3.1$, we have $\pi^2 > 9.61$, so the inequality holds. Hence for $\alpha = \pi/4$ (and, by continuity, for all α close to $\pi/4$) the circles cover more than four fifths of the triangle.

For the record, the best possible α comes from maximising $f(c) = (2-c^2)c$ with $c = \cos\alpha$: $f'(c) = 2 - 3c^2 = 0$ gives $\cos\alpha = \sqrt{2/3}$, i.e. $\sin\alpha = 1/\sqrt{3}$, and the maximum ratio is $\pi\sqrt{6}/9 \approx 0.855$.

Checks

- As $\alpha \to \pi/2$ the triangle flattens and $k \to 0$: the circles C_n vanish and $S \to \frac{1}{2}\pi r_0^2$, matching $\frac{1+1}{4}\pi r_0^2$.
- With $\alpha = \pi/6$, $k = \frac{1}{3}$ and $S = \pi r_0^2\left(\frac{1}{2} + \frac{1/9}{8/9}\right) = \frac{5}{8}\pi r_0^2$; the formula gives $\frac{1+1/4}{2}\pi r_0^2 = \frac{5}{8}\pi r_0^2$.
- Numerically, building 3000 circles by direct tangency for random α reproduces both formulae to about 10^{-15}, and the ratio at $\alpha = \pi/4$ is 0.8330.

Common slips

- Counting C_0 as a full circle: it is a semicircle, which is why the $\frac{1}{2}$ appears.
- Using $d_n + d_{n+1} = r_n + r_{n+1}$ or $d_n - d_{n+1} = r_n - r_{n+1}$: the circles touch externally and both centres are on the same side of O.
- Trying a small α for the last part: the ratio tends to $\pi/4 < \frac{4}{5}$ there.

Beyond STEP

The chain of circles is generated by a single dilation (enlargement) centred at O with scale factor k: it maps the corner region to itself and each circle to the next. Self-similar configurations like this lead to fractals and to the Apollonian circle packings, where the curvatures of mutually tangent circles obey Descartes' circle theorem. Inversion, a transformation that turns circles into lines, makes many such tangency problems easy; Needham's *Visual Complex Analysis* treats it beautifully.

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